# Homework — 1.3.4 Web Technologies

Name: ____ Date: _ ***Mark:*** ___ / 20

*OCR H446 — set after the 1.3.4 lesson. Show your working.*

## Part A — This week's topic (~14 marks)

**1.** A web page uses HTML, CSS and JavaScript together. A developer wants every paragraph styled with the same blue text, and wants one specific warning box styled differently.

(a) State which **CSS selector type** would be used to style **all** paragraphs the same way. **[1]** *(AO1)*

(b) State which **CSS selector type** (`.class` or `#id`) is most appropriate for styling the **single** warning box, and justify your choice. **[2]** *(AO2)*

**2.** Explain why **CSS** is used to control presentation rather than placing styling directly in the HTML of every page of a large website. **[2]** *(AO2)*

**3.** Describe the role of a **web crawler** in how a search engine builds its index. **[2]** *(AO1)*

**4.** The **PageRank** algorithm ranks pages by importance.

(a) Explain how an incoming link from a **high-ranking** page affects a page's rank differently from a link from a **low-ranking** page. **[2]** *(AO1)*

(b) State what is meant by saying PageRank is calculated **iteratively** until it **converges**. **[2]** *(AO1)*

**5.** A web form lets a user choose a username. The browser checks instantly that the box is not empty (client-side), and the server then checks that the username is not already taken in the database (server-side).

(a) Give **one** benefit of doing the "not empty" check **client-side**. **[1]** *(AO2)*

(b) Explain why the "username already taken" check **must** be done server-side. **[2]** *(AO2)*

## Part B — Retrieval: keep it fresh (~6 marks)

**6.** *(1.3.3)* State the difference between a **switch** and a **router**, referring to the type of address each uses. **[2]** *(AO1)*

**7.** *(1.3.1)* Explain **one** reason why **symmetric** encryption has a key distribution problem that **asymmetric** encryption avoids. **[2]** *(AO1)*

**8.** *(1.4.1)* Convert the denary number **45** into **8-bit binary**. Show your working. **[2]** *(AO2)*